

Integration Architecture

APIs, ESB, message queues, and data flows

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1. Introduction

This document describes the integration architecture that connects ArchiSurance's application components. It covers synchronous (REST API), asynchronous (event streaming), and batch integration patterns. The Application Portfolio (EA-APP-002) provides details on individual components; this document focuses on how they communicate.

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Lessons learned from the pilot phase have been incorporated into the current design iteration. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Stakeholders across multiple departments have contributed requirements for this workstream. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice.

The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings.

Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam.

A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The integration architecture patterns has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Lessons learned from the pilot phase have been incorporated into the current design iteration. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required.

2. Enterprise Service Bus

The Enterprise Service Bus is the backbone of ArchiSurance's integration layer. Built on Apache Kafka, it connects the Salesforce CRM, Billing Engine, and Policy Admin through message-based communication. The ESB handles message routing, transformation, and guaranteed delivery.

The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Security review was completed in accordance with the CISO's security-by-design framework. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Lessons learned from the pilot phase have been incorporated into the current design iteration. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema

validation, and audit logging applied at the gateway layer. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints.

Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour.

The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development.

ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes

and a post-incident review within five business days. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures.

3. API Gateway and REST APIs

The Kong API Gateway (exposed as the REST API Gateway interface) manages all external API traffic. It serves the Customer Self-Service Portal, Mobile App Backend, and Broker Portal. Rate limiting and caching are provided by Redis Cluster. Authentication is delegated to IAM.

A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Change management activities include stakeholder communication, training sessions, and documentation updates. Lessons learned from the pilot phase have been incorporated into the current design iteration. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards.

4. Event-Driven Integration (Kafka)

Event-driven integration is implemented through Apache Kafka with the following topic structure:

```
# Kafka Topic Registry (archisurance-prod)
claims.registered      - New claim events (producer: Claims Platform)
claims.assessed        - Assessment outcome (producer: Claims Platform)
```

fraud.scored	- Fraud score results (producer: Fraud Engine)
policy.issued	- New policy events (producer: PAS)
policy.renewed	- Renewal events (producer: PAS)
customer.onboarded	- New customer events (producer: CRM)
payment.processed	- Payment confirmations (producer: Billing)
notification.requested	- Notification triggers (consumer: Notification Svc)
analytics.events	- All events for DWH (consumer: Batch Processing)

The Batch Processing Engine consumes all topics and loads data into the Oracle DWH. The Notification Service subscribes to claim and policy events. The FDE consumes claim events and produces fraud scores consumed by the Claims Platform.

A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The vendor selection process followed the procurement guidelines established by the CFO's office. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures.

Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The information security team requires that all new

systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows.

The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The vendor selection process followed the procurement guidelines established by the CFO's office. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Lessons learned from the pilot phase have been incorporated into the current design iteration. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures.

5. Application Interface Catalog

REST API Gateway

The REST API Gateway provides external connectivity. All interfaces use TLS 1.3 and require OAuth 2.0 tokens issued by the Access Management System platform.

Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap.

Mobile App Interface

The Mobile App Interface provides external connectivity. All interfaces use TLS 1.3 and require OAuth 2.0 tokens issued by the Identity & Access Management platform.

Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys.

Partner Integration Hub

The Partner Integration Hub provides external connectivity. All interfaces use TLS 1.3 and require OAuth 2.0 tokens issued by the IAM platform.

The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline.

SOAP Legacy Interface

The SOAP Legacy Interface provides external connectivity. All interfaces use TLS 1.3 and require OAuth 2.0 tokens issued by the IAM platform.

The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024.

GraphQL Analytics Endpoint

The GraphQL Analytics Endpoint provides external connectivity. All interfaces use TLS 1.3 and require OAuth 2.0 tokens issued by the IAM platform.

Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points.

6. Data Flow Diagrams

The following application services define the data flow contracts:

- Policy API -- realized by Policy Admin
- Claims API -- realized by Claims Platform
- Risk Scoring API -- realized by Actuarial Risk Module
- Customer API -- realized by Customer Management Platform
- Notification API -- realized by Messaging Service
- Document API -- realized by Document Repository
- Fraud Scoring API -- realized by Fraud Detection Engine
- Payment API -- realized by Billing System

- Reporting API -- realized by BI Platform
- Authentication API -- realized by Access Management System

The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Lessons learned from the pilot phase have been incorporated into the current design iteration. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Security review was completed in accordance with the CISO's security-by-design framework. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model.

7. API Endpoint Reference

Claims API - OpenAPI Excerpt

```
openapi: 3.0.3
info:
  title: ArchiSurance Claims API
  version: 2.4.1
paths:
  /api/v2/claims:
    post:
      summary: Register a new claim (FNOL)
      operationId: registerClaim
      requestBody:
        required: true
        content:
```

```

    application/json:
      schema:
        $ref: '#/components/schemas/ClaimRequest'
      responses:
        '201':
          description: Claim registered
        '400':
          description: Validation error
/api/v2/claims/{claimId}:
  get:
    summary: Retrieve claim details
    parameters:
      - name: claimId
        in: path
        required: true
        schema:
          type: string
    responses:
      '200':
        description: Claim details
      '404':
        description: Claim not found

```

The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. Stakeholders across multiple departments have contributed requirements for this workstream. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility.

The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Stakeholders across multiple departments have contributed requirements for this workstream. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within

twenty-four hours upon regulatory request. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation.

8. Appendices

References and Related Documents

Doc ID	Title	Version
EA-OVR-001	Enterprise Architecture Overview	3.0
EA-APP-002	Application Portfolio & Inventory	4.2
EA-INFRA-003	Infrastructure Design Document	2.5
EA-BIZ-004	CRM Business Process Description	2.0
EA-CLM-005	Claims Processing Workflow	3.1
EA-SEC-006	Security Architecture	2.3
EA-INT-007	Integration Architecture	2.8
EA-DAT-008	Data Architecture & Governance	2.0
EA-CLD-009	Cloud Migration Assessment	1.5
EA-DR-010	Disaster Recovery & Business Contin	3.0
EA-NET-011	Network Architecture	2.2
EA-ROAD-012	Strategic IT Roadmap 2025-2027	2.0
EA-SVC-013	IT Service Catalog	3.5
EA-VND-014	Vendor & Technology Assessment	1.8
EA-OPS-015	Operational Runbook: Core Platform	4.1
EA-CMP-016	Compliance & Regulatory Requirement	2.5
EA-CHG-017	Change Management: ERP Migration	1.2

Distribution List

Name	Role	Responsibility
J. van den Berg	Enterprise Architect	Review & Approve
M. de Vries	Application Architect	Review & Contribute
P. Jansen	Integration Architect	Review & Contribute
K. Bakker	Infrastructure Architect	Review & Contribute
S. Visser	Business Analyst	Inform
T. Mulder	Process Architect	Inform
R. de Jong	CISO	Review & Approve
L. Hendriks	Data Architect	Inform
A. Willems	CTO	Final Approval
B. Smit	Head of Claims	Business Sign-off

Glossary of Acronyms

Acronym	Definition
API	Application Programming Interface
BPM	Business Process Management
CDN	Content Delivery Network
CISO	Chief Information Security Officer
CRM	Customer Relationship Management
DR	Disaster Recovery
DMS	Document Management System
DMZ	Demilitarized Zone
EA	Enterprise Architecture
ESB	Enterprise Service Bus
ETL	Extract, Transform, Load

FNOL	First Notice of Loss
GDPR	General Data Protection Regulation
HSM	Hardware Security Module
IAM	Identity and Access Management
KPI	Key Performance Indicator
LAN	Local Area Network
MTTR	Mean Time to Recovery
NPS	Net Promoter Score
PAS	Policy Administration System
PCI-DSS	Payment Card Industry Data Security Standard
PoC	Proof of Concept
RBAC	Role-Based Access Control
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SAN	Storage Area Network
SLA	Service Level Agreement
SOX	Sarbanes-Oxley Act
STP	Straight-Through Processing
VLAN	Virtual Local Area Network
VPN	Virtual Private Network
WAN	Wide Area Network

Appendix: Meeting Minutes - Architecture Review Board

Date: March 15, 2025. Location: ArchiSurance HQ, Room 4.12. Attendees: J. van den Berg (chair), M. de Vries, P. Jansen, K. Bakker, S. Visser, T. Mulder (remote). Apologies: R. de Jong (on leave).

Agenda item 1: Review of open action items from previous meeting. Action 2024-47 (Jansen): Provide updated cost estimate for Oracle migration. Status: Completed. The revised estimate is EUR 1.4M including contingency. Action 2024-48 (de Vries): Schedule penetration test with external vendor. Status: In progress, vendor shortlist reduced to two candidates.

Agenda item 2: Discussion on coffee machine replacement in the 3rd floor kitchen. The current Jura E8 has been experiencing frequent bean grinder failures. Facilities will procure a replacement by end of month. Not related to architecture but raised due to impact on team productivity during long review sessions.

Agenda item 3: Parking lot discussion about whether to adopt Backstage as the internal developer portal. Deferred to next meeting pending PoC results. K. Bakker noted that the onboarding documentation is outdated and should be refreshed regardless of the portal decision.

Next meeting: April 12, 2025 at 14:00 CET. Van den Berg to circulate agenda by April 5.

Detailed Revision Notes for Version 2.1

Section 2.3 was updated to reflect the new naming convention adopted by the Platform Engineering team. All references to 'microservice cluster' have been replaced with 'application workload group' per the Architecture Board decision of January 2025. Section 4.1 was expanded to include the results of the capacity planning exercise conducted in Q4 2024, which identified the need for additional compute resources in the Amsterdam primary data center.

Appendix B was added to include the RACI matrix requested by the PMO during the Q3 2024 quarterly review. The glossary was updated to include twelve new acronyms that were introduced since the previous version. Figure 3 was replaced with an updated diagram generated from the Sparx EA repository on March 28, 2025. The document classification was elevated from 'Internal' to 'Internal - Confidential' following the data classification review of February 2025.

Feedback from the peer review conducted by M. de Vries on March 10, 2025, addressed three items: (1) clarification of the boundary between the DMZ and the application tier, (2) addition of the backup retention policy reference, and (3)

correction of the SLA target for the Claims Handling Service from 99.5% to 99.9% as agreed in the contract amendment of December 2024.

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